



Cluster Robust Variance Estimation

Edoardo Di Cosimo



Introduction

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- The variance of the sum of 2 random variables X_1 and X_2 is:

$$\text{Var}(X_1) + \text{Var}(X_2) + 2 \cdot \underbrace{\text{cov}(X_1, X_2)}_{=0 \text{ if } X_1 \perp X_2}$$

- The OLS estimator can be written as the true parameter plus a sample (weighted) average of individual scores

$$\hat{\beta}_{\text{ols}} = \beta + \frac{\sum_i X_i u_i}{\sum_i X_i^2}$$



Asymptotic

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- To conduct hypothesis tests we need the asymptotic distribution of our estimators
- I.e. we need to understand how the estimator would behave under repeated sampling.

$$\sqrt{N} (\hat{\beta}_{\text{ols}} - \beta) = \frac{\frac{1}{\sqrt{N}} \sum_i X_i u_i}{\frac{1}{N} \sum_i X_i^2}$$



- Focus on the numerator

$$\frac{1}{\sqrt{N}} \sum_i X_i u_i$$

- This is a sum of N random variables
- If they are all independent:

$$\xrightarrow{d} \mathcal{N} \left(0, \frac{1}{N} \sum_i \text{Var}(X_i u_i) \right)$$



Which is the problem?

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- Inside a cluster the scores of 2 individuals are not independent
- Usual White Variance estimator is not consistent

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i \xrightarrow{d} \mathcal{N} \left(0, \frac{1}{N} \sum_{i=1}^N \text{var}(X_i u_i) + \frac{2}{N} \sum_{i \neq j}^N \sum_j^N \text{cov}(X_i u_i, X_j u_j) \right)$$

